

Sanath Palakkal

📍 Kozhikode

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in [sanathpalakkale](#)

Profile

Motivated M.Sc. student specializing in VLSI at Digital University Kerala with hands-on experience in analog circuit design, Verilog HDL, and CMOS layout. Strong interest in IC design, simulation, and embedded systems. Seeking an internship opportunity to apply theoretical knowledge to real-world semiconductor and hardware design challenges.

Domain skills

- **VLSI / IC Design Domain Skills :** CMOS circuit design fundamentals, Digital IC design basics, VLSI fabrication & technology understanding, Physical design basics (layout, DRC, LVS)
- **EDA Tools :** Verilog (RTL design), Cadence Virtuoso, LTspice (circuit simulation), Magic VLSI (layout Xschem (schematic capture), Silvaco TCAD (device-level simulation)
- **PCB & ECAD Tools :** KiCAD, Eagle
- **Embedded & Programming:** Arduino, C, Python, Verilog

Education

MSc	Digital University of Kerala, Trivandrum , MSc Applied physics with specialization in VLSI • Coursework: VLSI Technologies, VLSI Physical design, Digital IC design, Mixed signal design	2024 – Ongoing
BSc	Calicut University BSc Physics	2020 – 2023
HSS	GGVHSS Feroke Kerala , India Biology Science	2018 – 2020

Academic Projects

VGG-11 Neural Network Hardware Implementation

- Implemented VGG-11 architecture using **Verilog HDL**
- Performed behavioral simulation and hardware-oriented optimization
- Design targeted for **ASIC tape-out**
- Analyzed layer-wise operations and resource efficiency

Autonomous Object Detection Car using ESP32

- Developed a robot car using ultrasonic sensors and L293D motor driver.
- Wrote microcontroller code for obstacle detection and real-time response

Certificate

- Attended a Short-Term Training Programme on Custom IC Design and Layout using Standard 180nmPDK organized by National Institute of Technology, Calicut with IEEE SSCS.
- **VLSID 2026 Fellowship Recipient** – Selected for the fellowship at the VLSI Design Conference (VLSID 2026), held in **Pune, India**, in recognition of academic merit and interest in VLSI research

Area of Interest

Digital IC Design , Analog Circuit design , Verilog Programming , Verification & Validation

